

Multivariate Outlier Detection

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Winter 2026



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Recap

- Last time we covered **Distance Measures II**:
 - Euclidean distance.
 - Manhattan distance.
 - Jaccard similarity & dissimilarity.
 - Minkowski distance

Introduction

- We have covered some ways to identify outliers in certain scenarios.
 - Univariate: $\pm 1.5 \times IQR$ (boxplots)
 - Regression: Cook's distance.
- Cook's distance is based on how much influence each observation has on the linear regression model estimates.
- These measures apply to very specific scenarios.
 - *Are there general approaches?*

Example 1

- Import the *Outliers.csv* file and do the following:
 - ① Generate boxplots for each of the observations.
 - ② Construct a simple linear regression model each pair of X_i and Y_i .
 - ③ Construct the following multivariate linear regression model:
$$Y_1 \sim X_1 + X_2.$$
 - ④ Use Cook's distance to identify possible outliers in each of your three models.
 - ⑤ What did you notice?

Multivariate Normal Distribution

- A **multivariate normal distribution** is a generalization of the univariate normal distribution into multiple dimensions.
 - *It is a joint distribution of normal marginal distributions.*

$$\mathbf{X} \sim N_k(\mu, \Sigma).$$

- Where μ is a k -dimensional vector of expected (mean) values.
- And Σ is a $k \times k$ covariance matrix.
- *Time permitting, we will speak more about correlated joint distributions.*

Mahalanobis Distance

- **Mahalanobis distance** measures how likely an object belongs to a group of objects.
 - It is designed to work on multivariate normal data.
- In R:
 - `data$mahalanobis<- mahalanobis(data, colMeans(data), cov(data))`
 - `data$pvalue <- pchisq(data$mahalanobis, df=p-1, lower.tail=FALSE)`
 - *A p-value that is less than 0.001 suggests an outlier.*
- We may also use the *mvoutlier* package

Example 2

- Using the *Outliers.csv* file and do the following:
 - 1 Subset the data to only include Y_1 , X_1 , and X_2 .
 - 2 Calculate the Mahalanobis distance for each observation.
 - 3 According to a χ^2 -test are any of the observations outliers?
 - 4 Do you think that there are any potential problems with this?
 - 5 What did you notice?

Robust Mahalanobis Distance

- The classical Mahalanobis distance relies on all of the observations to estimate the means and covariances.
 - *Outliers may severely impact these estimates.*
- We can use the *central* part of the data to estimate these values.
- Using the *mvoutlier* package in R:
 - `aq.plot(x,delta=qchisq(0.975,df=ncol(x)),
 ,quan=1/2,alpha=0.05)`
 - `quan`= how much of the data you want to use for estimation [0.5 - 1].

Example 3

- Using the *Outliers.csv* file and do the following:
 - 1 Repeat your analysis from Example 2 using the `aq.plot()` function.
 - 2 Change the `quan= 90%` of the data.
 - 3 Compare your results to Example 2.
 - 4 What did you notice?

Non-Normal Data

- Outlier detection with large departures from a normal distribution relies heavily on the objective of your work.
- For example, if you were looking at multivariate data that contained salaries, you would expect *outliers* in the positive tail.
- We can rely on distance measures or clustering methods to identify these points.
 - *We will talk about clustering later.*

Heatmaps and Distance Measures

- We can combine heatmaps with distance measures to visually identify possible outliers.

- In R:

```
dist(Outliers2) %>%  
as.matrix() %>%  
heatmap()
```

- Darker values indicate a larger distance.
- *Also generates hierarchical clustering results.*

Example 4

- Using your dataset from Example 2 calculate the following distance matrices and generate their corresponding heatmaps:
 - 1 Euclidean distance.
 - 2 Manhattan distance.
 - 3 Minkowski distance ($p = 1.5$, $p = 3$, $p = 50$)
 - 4 What did you notice?

Comments on Outlier Detection

- Multivariate outlier detection relies heavily on the objective of your work.
 - There are many methods for univariate outlier detection.
- **Just because an object is flagged as an outlier, does not mean we can simply delete it.**
- *Outlier detection is an open area of scientific research.*

Next Lecture

- We will discuss **Clustering**.
- Hierarchical Clustering
- *K-Means Clustering*

Exercise 1

- Import the Football24.csv file and use the methods discussed in this lecture to determine if there are any potential outliers.
- What do you notice across the different methods?

Exercise 2

- Do you think we could use dissimilarity measures to identify possible outliers?
- How might you implement this?

References & Resources

- ① Bakker, J. D. (2026) Applied Multivariate Statistics in R
 - ② Peng R. D. (2020) *Exploratory Data Analysis with R*
 - ③ Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC
- `dist()`
 - `vegan::vegdist()`
 - `labdsv::dsvdis()`
 - `heatmap`
 - `mvoutlier`